

Sheet 2 - matrices

1a) $AB = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 4 & 3 \\ 5 & 6 & 5 \end{pmatrix}$

BC undefined

you mean 5 here?

$$AC = \begin{pmatrix} 4 & 4 \\ 11 & 9 \\ 17 & 12 \end{pmatrix}$$

1a. otherwise, correct

CA undefined.

A+B undef.

$$A+B' = \begin{pmatrix} 2 & 2 \\ 3 & 5 \\ 6 & 6 \end{pmatrix}$$

b) $C'A' = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 11 & 17 \\ 4 & 8 & 12 \end{pmatrix} = (AC)'$ from (a)

1b. correct

c) $C^{-1} = \frac{1}{|C|} \cdot \begin{pmatrix} 2 & 0 \\ -2 & 1 \end{pmatrix}$ where $|C| = 2$.

1c. ok, correct

$$(C')^{-1} = \frac{1}{2} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}^{-1} = \frac{1}{2} \cdot \begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix} = (C^{-1})'$$

d) $C'C = \begin{pmatrix} 5 & 4 \\ 4 & 4 \end{pmatrix}$ $(C'C)^{-1} = \frac{1}{4} \cdot \begin{pmatrix} 4 & -4 \\ -4 & 5 \end{pmatrix}$

1d. yes, correct

$$C^{-1}(C')^{-1} = \frac{1}{2} \cdot \frac{1}{2} \cdot \begin{pmatrix} 2 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix} = \frac{1}{4} \cdot \begin{pmatrix} 4 & -4 \\ -4 & 5 \end{pmatrix}$$

2. a) $\text{tr}(A) = 9$, $\text{tr}(B) = 3$.

2.

~~2A~~ minors from A:

$$\begin{pmatrix} 8 & 6 & 0 \\ -2 & 4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

so $|A| = 2 \cdot 8 - 1 \cdot 6 = 10$

so inverse is

$$\begin{pmatrix} 8 & 6 & 0 \\ -2 & 4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

For B, consider bottom row, with cofactors $(-5 \quad +19 \quad -8)$
then $|B| = -8$.

$$AB = \begin{pmatrix} 9 & -1 & 8 \\ 31 & 1 & 17 \\ 0 & 0 & 2 \end{pmatrix}$$

$$BA = \begin{pmatrix} -1 & -3 & 6 \\ 17 & 11 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

$$|A B| = 2 \times 40 = 80 = |A| |B|$$

2. ok, all correct!

$$\text{tr}(AB) = 12 = \text{tr}(BA)$$

3. $X_{(m \times n)}$ is rank n , with $m \geq n$. (i.e. full-rank)

a) $[X'X \text{ has dimensions } n \times n]$
 $X'X$ is true definite if its quadratic form > 0 for all nonzero a .
 So consider $M = a' X' X a$, where a is a nonzero vector.

We have $M = (Xa)' Xa$ by properties of transpose. Letting $W = Xa$, note that $M = W'W$. Since $(W'W)' = W'W'' = W'W$, M is symmetric.

3a. ok, correct!

Note that W will be an $m \times 1$ column vector. So $(W)'W$ is the dot product of W with itself, i.e. the sum of squares of all its elements, which must always be nonnegative. Given a is nonzero, we also have that W is nonzero from the fact that X is full rank (i.e. it won't map any vectors to the zero vector besides the zero vector itself).

So M is always positive $\Rightarrow X'X$ is true definite $\Rightarrow$ it is invertible i.e. non-singular.

b) $P = X(X'X)^{-1}X'$ is an $m \times m$ matrix, as is M .

$$PP = (X(X'X)^{-1}X')(X(X'X)^{-1}X') = (X(X'X)^{-1}(X'X)(X'X)^{-1}X') \text{ by associativity} \\ = (X(X'X)^{-1}I X') = X(X'X)^{-1}X' = P \therefore \text{ idempotent.}$$

$$P' = (X(X'X)^{-1}X')' = X''((X'X)^{-1})'X' = X(X'X)^{-1}X' \text{ since } X'X \text{ is symmetric} \\ = P \text{ so symmetric}$$

$$MM = I^2 - IP - PI + P^2 \text{ by distributive property, all } m \times m \text{ matrices} \\ = I - P - P + P = I - P = M \text{ since } P \text{ idempotent}$$

3b. yes, correct

$$M' = I' - P' \quad \text{because transpose is a linear operation} \\ = I - P = M \quad \text{by symmetry, so } M \text{ also symmetric}$$

$$c) \operatorname{tr}(P) = \operatorname{tr}(X'X(X'X)^{-1}) \quad \text{by cyclic property} \\ = \operatorname{tr}(I_n) = n \quad \text{since } X'X \text{ is } n \times n.$$

$$\operatorname{tr}(M) = \operatorname{tr}(I_m) - \operatorname{tr}(P) \quad \text{by linearity} \\ = m - n \quad \text{by above}$$

3c. ok, correct

$$d) X'(y - P_y) = X'y - X'P_y \quad \text{by distr.} \\ = X'y - X'X(X'X)^{-1}X'y \\ = X'y - I X'y \quad \text{by assoc.} \\ = 0 \quad \text{i.e. the residual } M_y \text{ is orthogonal to } X.$$

$P_y = X(X'X)^{-1}X'y$. It projects y into the column space of X .

$(\Rightarrow)$ Suppose $y = Xc$ for some $n \times 1$ vector c .

$$\text{Then } P_y = X(X'X)^{-1}X'Xc = XIc = Xc = y.$$

$(\Leftarrow)$ Suppose $P_y = y = X(X'X)^{-1}X'y$

Then let $c := (X'X)^{-1}X'y$ is an $n \times 1$ vector, and

$$y = Xc$$

3d. yes, well done!

Our prediction of the independent variable y is equal to its actual value only if y is in the column space of regressors, i.e. if the errors u are zero.

$$e) X'X = 2, \quad (X'X)^{-1} = \frac{1}{2} \quad \therefore P = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_y = \frac{1}{2} \begin{pmatrix} y_1 + y_2 \\ y_1 + y_2 \end{pmatrix} \\ \text{where } y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$M = I - P = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \text{ is symmetric, as is } P \text{ above.}$$

could overlap

4. Yes. Informally: let $A = (1, 2, 0)$, $B = (-1, 0, 0)$, $C = (1, 1, 1)$. Note that only C has a z component, so it can't be involved in a combination summing to zero. But of $\{A, B\}$ only A has a y component, so no combination can sum to zero, since they both have z components.

Q4. yes, correct!

Formally, consider $\det \begin{pmatrix} 1 & 2 & 0 \\ -1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} = \begin{vmatrix} 1 & 2 \\ -1 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ 0 & 0 \end{vmatrix} = 2 + 0 + 0 = 2 \neq 0$

so the square matrix composed of them is invertible and thus full-rank, so they're independent.

Q5. yes, correct!

5. $\text{rank}(B) \stackrel{!}{=} 2$ since there are two linearly independent rows (and rank is at most $\min(n\text{-row}, n\text{-col})$)

$\text{rank}(C) = 1$ since all rows are linear combinations of each other.

4e) $P^2 = \frac{1}{4} \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = P$; $M^2 = \frac{1}{4} \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = M$
 cont'd. $\checkmark(b)$

$\text{tr}(P) = 1$, $\text{tr}(M) = 1$ with $m=2, n=1$ $\checkmark(c)$.

(Not sure why finding P_y was relevant?)

This is a constant-only regression with two observations. Note that our fitted values are — as expected — the sample mean $\frac{1}{2}(y_1 + y_2)$ for each $m=2$ observations.

3e. yes, all correct! P is the projection matrix, it contains the "predicted values" of y and is geometrically the point "closest" to y in X -space. In this special case, our best prediction of y is the simple average of its elements. Our prediction would be perfect (no error) if both elements of y were the same i.e., if it were $(5, 5)$, $(4, 4)$.